MD. NAHIN ALAM

AI Automation Engineer · Full-Stack Architect · AI/ML Researcher

North South University | Dhaka, Bangladesh

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RESEARCH PROFILE

Results-driven AI Automation Engineer and Full-Stack Architect with deep expertise in multimodal AI, medical imaging, and under-resourced language processing. Specialises in designing production-ready LLM ecosystems, scalable RAG pipelines, and enterprise SaaS architectures. Active AI/ML researcher affiliated with North South University and the Centre for UntieAI Artificial Intelligence Research Lab at CU Boulder, with a publication record spanning top venues including *KDD, ACL, MICCAI, SaTC, ICMLAS, Health*, and Plos One. Has co-authored 10+ research papers; processed 500 GB+ of complex multimodal datasets; reduced manual data processing time by 60%+; and achieved 99.9% uptime on multi-tenant cloud deployments.

Research Interests: Multimodal AI · Medical Imaging & Radiology · Vision-Language Models · NLP for Under-resourced Languages · IoT & Predictive Maintenance

EDUCATION

Bachelor of Science — Computer Science & Engineering *Jan 2020 – Apr 2024*

North South University, Dhaka, Bangladesh

CGPA: 3.81 / 4.00 (96th percentile) | **Honors:** Summa Cum Laude, Dean's List, EQF Level 6

English Proficiency: IELTS 6.5 Overall (L:6.5 / R:6.5 / W:6.5 / S:6.5) — Sep 2025 | TRF: 25BD507770ALAM001A

Higher Secondary Certificate (HSC) *2017 – 2019*

Uttara High School & College, Dhaka, Bangladesh

Result: A Grade / First Class (77%)

Secondary School Certificate (SSC) *2015 – 2017*

Uttara High School & College, Dhaka, Bangladesh

Result: A* / High First Class (80–100%)

PROFESSIONAL & RESEARCH EXPERIENCE

AI & Automation Engineer *Jan 2026 – Present*

MySoftHaven (BD) Ltd., Dhaka, Bangladesh

- Develop AI software solutions for government-level projects, encompassing end-to-end LLM integration, RAG pipelines, and workflow automation.
- Architect production-ready AI ecosystems using FastAPI, FAISS, Google Gemini 1.5 Flash, and Docker, reducing manual data processing time by 60%+.
- Design multi-tenant SaaS architectures with automated CRM sync concepts for Salesforce and HubSpot integration.

AI Research Assistant (Remote) *Jan 2025 – Present*

Centre for UntieAI Artificial Intelligence Research Lab, University of Colorado Boulder

- Constructed automated data pipelines to process 500 GB+ of complex multimodal medical imaging and NLP datasets, decreasing data preparation time by 50%.
- Co-authored 3 high-impact research papers currently under review at top-tier venues (KDD, ACL, SaTC) by driving experimental design and scalable AI methodologies.

Lab Instructor — AI/ML *Feb 2025 – Present*

Department of CSE, North South University, Dhaka

- Instruct core AI/ML laboratory sessions for 50+ students per semester, delivering hands-on PyTorch and TensorFlow demonstrations that improved average class performance by 15%.

- Engineered an automated Python-based grading system, reducing assignment evaluation time by 80% while providing instant, accurate feedback.
- Teaching scope: Data Structures & Algorithms, Object-Oriented Programming, C++, Java.

Vision-Language Model Researcher *Jan 2025 – Oct 2025*

Machine Learning Lab, North South University, Dhaka

- Spearheaded benchmarking of state-of-the-art VLMs (CLIP, ViLT, BioViL-T) across 10,000+ data points to identify architectural bottlenecks in frozen models for radiology tasks.
- Developed RadMultiBench — automated evaluation scripts accelerating multimodal medical model testing cycles by 3×.
- Research areas: Vision-Language Models, Radiology Benchmarks, Multimodal Hate Detection, Reverse Digital Footprint Search.

Cloud Infrastructure & Automation Specialist *Jun 2024 – Oct 2025*

The Data Island, Dhaka, Bangladesh

- Architected and deployed scalable multi-tenant cloud infrastructure across AWS and GCP using Docker and Kubernetes, achieving 99.9% system availability.
- Accelerated deployment cycles by 40% by engineering automated CI/CD pipelines for enterprise AI applications.
- Optimised REST APIs and high-availability PostgreSQL databases, reducing average query response time by 35% and enabling 5,000+ concurrent requests.

Teaching Assistant *Jun 2024 – Jan 2025*

Department of CSE, North South University, Dhaka

- Assisted faculty with undergraduate courses in Python, Java, and C/C++; supervised labs and conducted sessions on algorithms and data structures.

Research Assistant — Cyber Physical Systems Lab *Jun 2023 – Jun 2024*

North South University, Dhaka

- Developed intelligent sensor networks and implemented ML algorithms for predictive maintenance in IoT systems using TensorFlow and MATLAB.
- Focused on embedded systems, real-time data acquisition, and spatiotemporal forecasting.

PUBLICATIONS

ACCEPTED

From Benchmarks to Deployment: A Calibrated, SHAP-Guided, Production-Ready Pipeline for Diabetes Risk Prediction
Md. Nahin Alam et al. (First Author)
28th International Conference on Computer and Information Technology (ICCIT 2025) · **Accepted**

Validating a Consensus Feature Selection Pipeline for Rheumatoid Arthritis Biomarkers: A Synthetic Data Study
Md. Nahin Alam et al. (First Author)
ICMLAS 2026 · **Accepted**

Engineering Spatiotemporal Forecasts at Scale: Dhaka Weather with State-of-the-Art Machine Learning Models
Co-author
SaTC 2026 · **Accepted**

Harnessing Vision Language Models for Negation Sensitivity in Visual Question Answering
Co-author
SaTC 2026 · **Accepted**

UNDER REVIEW / SUBMITTED

RadMultiBench: A Multimodal Benchmark Revealing the Limits of Frozen Vision-Language Models in Radiology
Md. Nahin Alam et al. (First Author)
Health (Journal) — 2025 · **Under Review**
Paper & Code: researchgate.net | github.com/alamnahin/RadMultiBench

A Leakage-Safe Multimodal Medical Generalist Benchmark: Chest X-Ray Classification and Image-to-Impression Generation under Natural Imbalance
Author team
MICCAI 2026 · **Submitted**

Bridging the Gap in Low-Resource Languages: Multilingual Multimodal Hate Detection in Memes with Image Descriptions
Md. Nahin Alam (First Author)
Plos One · **Under Review**

Face2Profile: A Face-and-URL Dataset for Open-Ended Profile Construction

Co-author
KDD 2026 — Datasets & Benchmark Track · **Under Review**

The Differential Sensitivity Fusion Approach to Layerwise DCNN Pruning
Author team
Plos One 2026 · **Submitted**

Decentralizing Reputation Scoring Using Blockchain Behavioral Data and Machine Learning
Author team
MMAR-2026 · **Submitted**

Color-Aware Swin Transformers for Tea Leaf Disease and Pest Classification
Co-author
Artificial Intelligence in Agriculture (Journal) · **Under Review**

TECHNICAL SKILLS

AI / ML Frameworks	PyTorch · TensorFlow · Hugging Face Transformers · Scikit-learn · XGBoost · LightGBM · FastAI
Vision-Language Models	CLIP · ViLT · BLIP · BioViL-T · Google Gemini · LLM Orchestration · Multimodal Pipelines
LLMOps & RAG	RAG Architecture · FAISS Vector Databases · Google Genkit · Semantic Search · Multi-agent Systems
Backend & Cloud	FastAPI · Node.js · AWS · GCP · Docker · Kubernetes · CI/CD (GitHub Actions) · PostgreSQL · Firebase · REST API
Frontend	Next.js 15 · React 19 · TypeScript · Tailwind CSS v4 · Shadcn UI · TanStack Start · Prisma ORM
Languages	Python · TypeScript · C/C++ · Java · SQL / NoSQL
Tools & Platforms	MATLAB · Supabase · Strapi CMS · Redux Toolkit · Socket.io · n8n Automation · Recharts · Zod

SELECTED PROJECTS

MySoftHeaven AI Suite
FastAPI · FAISS · RAG · Gemini 1.5 Flash · Docker
Production-ready AI ecosystem with RAG chatbot, FAISS vector indexing, multi-tenant architecture, and automated CRM sync concepts for Salesforce and HubSpot.

CodeGenius — AI Learning Platform
Next.js 15 · React 19 · Google Genkit · Gemini 2.5 Flash · Firebase
Personalized coding education platform generating custom learning paths and real-time code examples via structured Gemini outputs.

Duari — AI Study Abroad Companion
Next.js 15 · Genkit · Firebase · Prisma ORM · Recharts
Full-stack education platform generating personalized university recommendations using Gemini 2.5 Flash with drag-and-drop tracking and data visualization.

Nexus — AI Knowledge Cartographer
Gemini 2.5 Flash · React/TypeScript · SVG Visualizer
Interactive research assistant that visualizes complex topics as dynamic knowledge graphs using structured JSON outputs from Gemini API.

RBAC Project Management System
PERN Stack · TypeScript · Prisma ORM · Redux Toolkit · JWT
Enterprise-grade administration platform with granular role-based access control, cryptographically secure invitation workflows, and soft-delete mechanisms.

AntHive — Real-time Language Exchange
MERN · Stream Chat SDK · Stream Video SDK · Zustand · TanStack Query
Comprehensive social application with high-performance real-time messaging and video conferencing.

AI-Powered Resume Builder
React 18 · Vite · Gemini AI · Strapi CMS · Clerk Auth
Full-stack SaaS with contextual AI-generated summaries, live preview, and one-click PDF export.

CodePrep — Interview Platform
Next.js 14 · Supabase · PostgreSQL · RLS · Server Actions
Algorithm practice platform with secure remote code execution sandbox and Row Level Security.

CERTIFICATIONS

- FastAI Deep Learning Course
- Stanford Machine Learning Specialization (Coursera) — Andrew Ng
- Andrew Ng Deep Learning Specialization (Coursera)
- freeCodeCamp Data Science Certification
- Google Cloud Platform Fundamentals

REFERENCES

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Personal Details: DOB: 19 May 2000 · Nationality: Bangladeshi · Passport: A15513857 (Valid to 28 Aug 2034) · Mother Tongue: Bengali · Fluent in English